DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

Project Title : Lenskart E-Commerce Order Management System

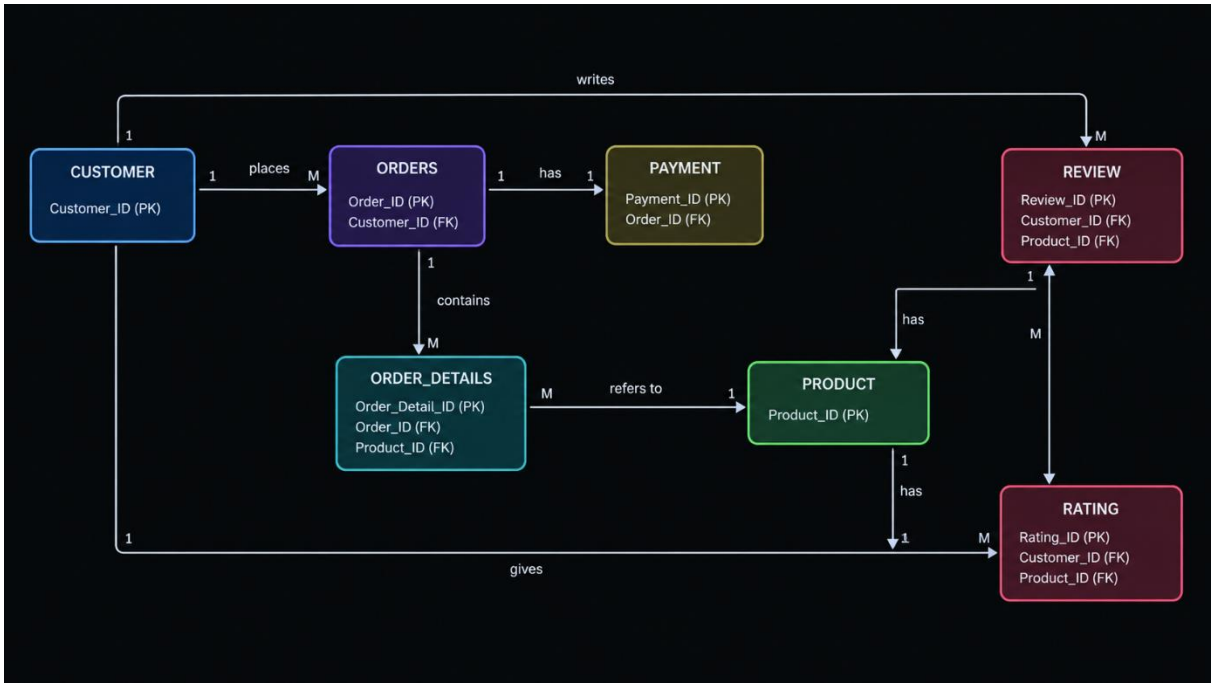
Objectives

1. Combine Customer, Product, Order, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. Combining Multiple Tables

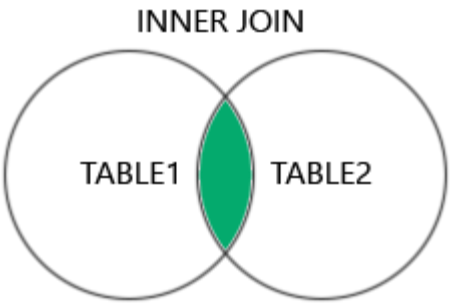
The Lenskart database contains different tables that store customer, product, Order, and payment information.

The tables can be combined using JOIN operation to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  INNER JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME
 ORDER_ID
----------- ------------------------------------
---------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Arun Kumar
       101
20-SEP-26         2499

          2 Priya
       102
21-SEP-26         1899

          3 Karthik
       103
22-SEP-26         2999


CUSTOMER_ID CUSTOMER_NAME
 ORDER_ID
```

This displays customer who have placed orders.

2.2 Orders and Payment

```
SQL> SELECT
  2      o.ORDER_ID,
  3      p.PAYMENT_ID,
  4      p.PAYMENT_STATUS,
  5      p.AMOUNT
  6  FROM ORDERS o
  7  INNER JOIN PAYMENT p
  8      ON o.ORDER_ID = p.ORDER_ID;

  ORDER_ID PAYMENT_ID PAYMENT_STATUS           AMOUNT
---------- ---------- -------------------- ----------
       101        201 Paid                       2499
       102        202 Paid                       1899
       103        203 Paid                       2999
       104        204 Paid                       1599
       105        205 Paid                       3499
       106        206 Paid                       1299

6 rows selected.
```

This displays order and corresponding payment details.

2.3 Product and Order_Details

```
SQL> SELECT
  2      p.PRODUCT_ID,
  3      p.PRODUCT_NAME,
  4      od.ORDER_ID,
  5      od.QUANTITY,
  6      od.PRICE
  7  FROM PRODUCT p
  8  INNER JOIN ORDER_ITEM od
  9      ON p.PRODUCT_ID = od.PRODUCT_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
---------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------
       201
Classic Eyeglasses
       101          2       1200

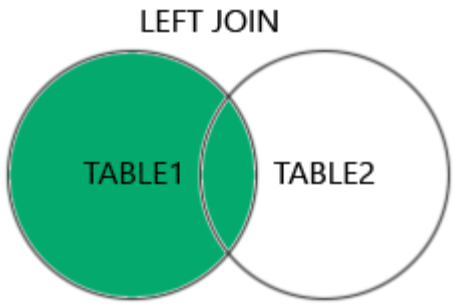
       202
Stylish Sunglasses
       102          1       1500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
---------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------
```

This displays products that have been included in orders.

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_STATUS,
  6      o.TOTAL_AMOUNT
  7  FROM CUSTOMER c
  8  LEFT JOIN ORDERS o
  9      ON c.CUSTOMER_ID = o.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ----------------------------------------------------
---------
ORDER_STATUS         TOTAL_AMOUNT
-------------------- ------------
          1 Arun Kumar
     101
Pending                      2499

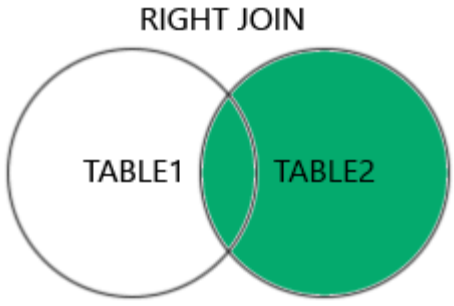
          2 Priya
     102
Processing                   1899

          3 Karthik
     103
Shipped                      2999


CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ----------------------------------------------------
```

This displays **all customers,** including customers who have not placed an order.

4. RIGHT JOIN



RIGHT JOIN returns all records from the right table and matching records from the left table.

4.1 Display All Orders and Customer Details

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_STATUS,
  6      o.TOTAL_AMOUNT
  7  FROM CUSTOMER c
  8  RIGHT JOIN ORDERS o
  9      ON c.CUSTOMER_ID = o.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- --------------------------------------------------
---------
ORDER_STATUS         TOTAL_AMOUNT
-------------------- ------------
          1 Arun Kumar
     101
Pending                      2499

          2 Priya
     102
Processing                   1899

          3 Karthik
     103
Shipped                      2999


CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- --------------------------------------------------
```

This displays **all orders** along with the matching customer information.

5.Complete Order Details

This query combines **Customer, Orders, Order_Details, Product, and Payment** tables.

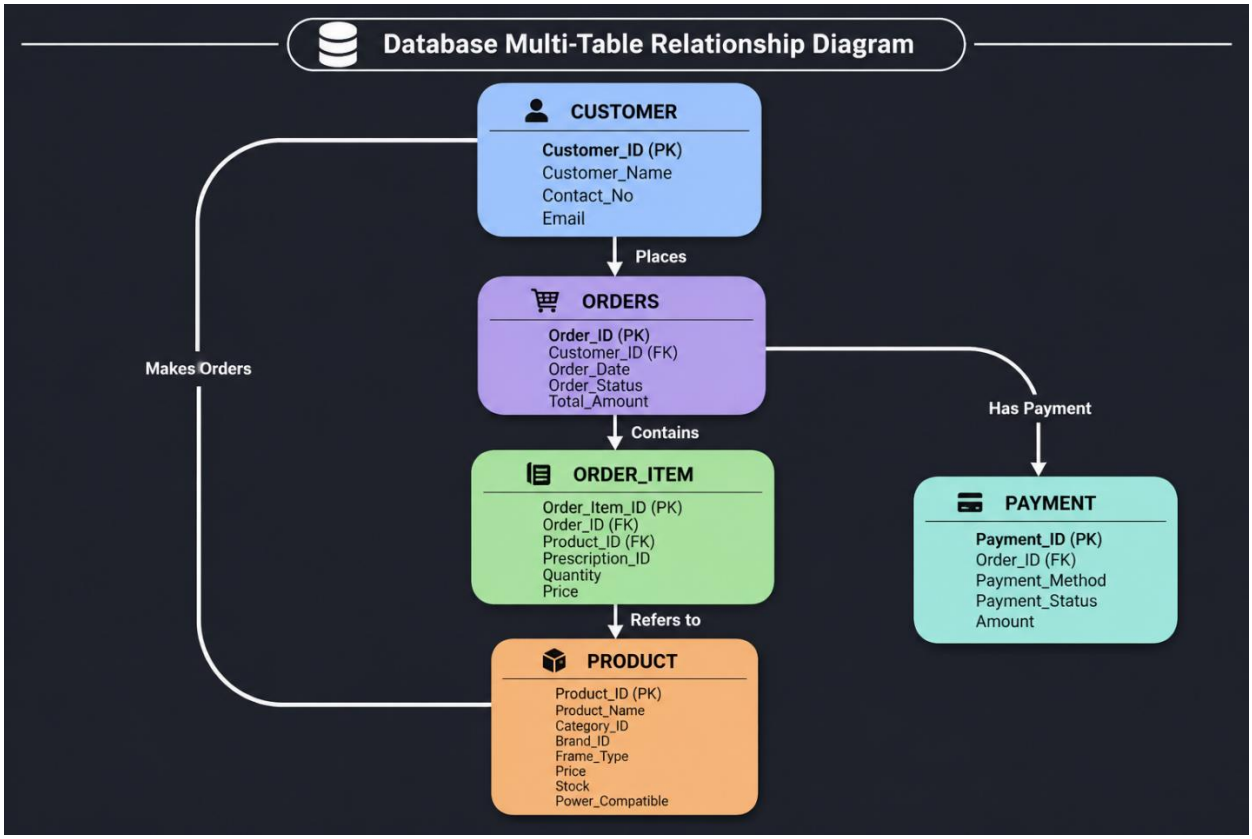
```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.ORDER_STATUS,
  7      p.PRODUCT_ID,
  8      p.PRODUCT_NAME,
  9      oi.QUANTITY,
 10      oi.PRICE,
 11      pay.PAYMENT_METHOD,
 12      pay.PAYMENT_STATUS,
 13      pay.AMOUNT
 14  FROM CUSTOMER c
 15  INNER JOIN ORDERS o
 16      ON c.CUSTOMER_ID = o.CUSTOMER_ID
 17  INNER JOIN ORDER_ITEM oi
 18      ON o.ORDER_ID = oi.ORDER_ID
 19  INNER JOIN PRODUCT p
 20      ON oi.PRODUCT_ID = p.PRODUCT_ID
 21  INNER JOIN PAYMENT pay
 22      ON o.ORDER_ID = pay.ORDER_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
ORDER_DAT ORDER_STATUS         PRODUCT_ID
--------- -------------------- ----------
PRODUCT_NAME
----------------------------------------------------------------
----------------
```

```
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
----------
         1 Arun Kumar
     101
20-SEP-26 Pending                     201
Classic Eyeglasses

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
ORDER_DAT ORDER_STATUS         PRODUCT_ID
--------- -------------------- ----------
PRODUCT_NAME
----------------------------------------------------------------
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
----------
         2       1200 UPI                         Paid
      2499


CUSTOMER_ID CUSTOMER_NAME
```



6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      p.PRODUCT_ID,
  6      p.PRODUCT_NAME,
  7      oi.QUANTITY,
  8      oi.PRICE
  9  FROM CUSTOMER c
 10  INNER JOIN ORDERS o
 11      ON c.CUSTOMER_ID = o.CUSTOMER_ID
 12  INNER JOIN ORDER_ITEM oi
 13      ON o.ORDER_ID = oi.ORDER_ID
 14  INNER JOIN PRODUCT p
 15      ON oi.PRODUCT_ID = p.PRODUCT_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
PRODUCT_ID
----------
PRODUCT_NAME
------------------------------------------------------------
----------------
  QUANTITY      PRICE
---------- ----------
          1 Arun Kumar
       101
         201
Classic Eyeglasses
         2       1200
```

7. Customer Purchase History with Payment

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      p.PRODUCT_NAME,
  6      oi.QUANTITY,
  7      oi.PRICE,
  8      pay.PAYMENT_METHOD,
  9      pay.PAYMENT_STATUS,
 10      pay.AMOUNT
 11  FROM CUSTOMER c
 12  INNER JOIN ORDERS o
 13      ON c.CUSTOMER_ID = o.CUSTOMER_ID
 14  INNER JOIN ORDER_ITEM oi
 15      ON o.ORDER_ID = oi.ORDER_ID
 16  INNER JOIN PRODUCT p
 17      ON oi.PRODUCT_ID = p.PRODUCT_ID
 18  INNER JOIN PAYMENT pay
 19      ON o.ORDER_ID = pay.ORDER_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
PRODUCT_NAME
----------------------------------------------------------------
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
```

```
          1 Arun Kumar
     101
Classic Eyeglasses
         2       1200 UPI                           Paid
      2499


CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
PRODUCT_NAME
----------------------------------------------------------------
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
----------
          6 Keerthana
     106
Classic Eyeglasses
         2       1200 Debit Card                    Paid
      1299


CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
PRODUCT_NAME
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.ORDER_STATUS,
  7      p.PRODUCT_NAME,
  8      oi.QUANTITY,
  9      oi.PRICE,
 10      pay.PAYMENT_METHOD,
 11      pay.PAYMENT_STATUS,
 12      pay.AMOUNT
 13  FROM CUSTOMER c
 14  INNER JOIN ORDERS o
 15      ON c.CUSTOMER_ID = o.CUSTOMER_ID
 16  INNER JOIN ORDER_ITEM oi
 17      ON o.ORDER_ID = oi.ORDER_ID
 18  INNER JOIN PRODUCT p
 19      ON oi.PRODUCT_ID = p.PRODUCT_ID
 20  INNER JOIN PAYMENT pay
 21      ON o.ORDER_ID = pay.ORDER_ID
 22  ORDER BY c.CUSTOMER_ID, o.ORDER_ID;

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
ORDER_DAT ORDER_STATUS
--------- --------------------
PRODUCT_NAME
----------------------------------------------------------------
----------------
```

```
PRODUCT_NAME
----------------------------------------------------------------
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
----------
         1 Arun Kumar
     101
20-SEP-26 Pending
Classic Eyeglasses

CUSTOMER_ID CUSTOMER_NAME
ORDER_ID
----------- ---------------------------------------------- -
---------
ORDER_DAT ORDER_STATUS
--------- --------------------
PRODUCT_NAME
----------------------------------------------------------------
----------------
  QUANTITY      PRICE PAYMENT_METHOD                PAYMENT_STA
TUS
---------- ---------- ---------------------------- -----------
---------
    AMOUNT
----------
         2       1200 UPI                          Paid
      2499
```

9. Payment Transaction Report

```
SQL> SELECT
  2      pay.PAYMENT_ID,
  3      o.ORDER_ID,
  4      c.CUSTOMER_ID,
  5      c.CUSTOMER_NAME,
  6      pay.PAYMENT_METHOD,
  7      pay.PAYMENT_STATUS,
  8      pay.AMOUNT,
  9      pay.PAYMENT_DATE
 10  FROM PAYMENT pay
 11  INNER JOIN ORDERS o
 12      ON pay.ORDER_ID = o.ORDER_ID
 13  INNER JOIN CUSTOMER c
 14      ON o.CUSTOMER_ID = c.CUSTOMER_ID
 15  ORDER BY pay.PAYMENT_ID;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT P
AYMENT_D
------------------------------ -------------------- ---------- -
--------
       201        101           1
Arun Kumar
UPI                            Paid                       2499 2
0-SEP-26

       202        102           2
Priya
Credit Card                    Paid                       1899 2
```

```
PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT P
AYMENT_D
------------------------------ -------------------- ---------- -
--------

       203        103           3
Karthik
Debit Card                     Paid                       2999 2
2-SEP-26

       204        104           4
Divya

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT P
AYMENT_D
------------------------------ -------------------- ---------- -
--------
Net Banking                    Paid                       1599 2
3-SEP-26

       205        105           5
Vignesh
UPI                            Paid                       3499 2
4-SEP-26
```